

SHETH L.U.J AND SIR M.V COLLEGE

Artificial Intelligence

Practical No.8

Aim : Association Rule Mining

- Implement the Association Rule Mining algorithm (e.g., Apriori) to find
- frequent itemsets.
- Generate association rules from the frequent itemsets and calculate their
- support and confidence.
- Interpret and analyze the discovered association rules.

Code:

```
Python 3.11.3 - C:\Users\Gazala\Documents\VS\SEM 5\AI\Rule Mining\3.11.3
File Edit Insert Run Options Window Help
import pandas as pd
import matplotlib.pyplot as plt

from mlxtend.frequent_patterns import apriori, association_rules

# Load the Superstore dataset
df = pd.read_csv('Superstore.csv', encoding='latin1')

# Display the first five records
print('First five records:')
print(df.head())

# Create transactions using Order ID and Sub-Category
basket = pd.crosstab(
    df['order_id'],
    df['Sub-Category']
)

# Convert the transaction values into True and False
basket = basket > 0

print('Transaction Data:')
print(basket.head())

# Find frequent itemsets using the Apriori algorithm
frequent_itemsets = apriori(
    basket,
    min_support=0.01,
    use_colnames=True
)

print('Frequent Itemsets:')
print(frequent_itemsets)

# Generate association rules using confidence
rules = association_rules(
    frequent_itemsets,
    metric='confidence',
    min_threshold=0.8
)

# Sort the rules according to confidence
rules = rules.sort_values(
    by='confidence',
    ascending=False
)
```

```
Python 3.11.3 - C:\Users\Gazala\Documents\VS\SEM 5\AI\Rule Mining\3.11.3
File Edit Insert Run Options Window Help

# Sort the rules according to confidence
rules = rules.sort_values(
    by='confidence',
    ascending=False
)

# Display support, confidence and lift
print('Association Rules:')
print(
    rules[
        ['antecedents',
         'consequents',
         'support',
         'confidence',
         'lift']
    ].head(10)
)

# Create readable names for the rules
top_rules = rules.head(10).copy()

top_rules['rule'] = (
    top_rules['antecedents'].apply(
        lambda x: ' '.join(list(x))
    )
    + ' => ' +
    top_rules['consequents'].apply(
        lambda x: ' '.join(list(x))
    )
)

# Visualize support and confidence of the top rules
plt.figure(figsize=(10, 6))

plt.bar(
    top_rules['rule'],
    top_rules['support'],
    label='Support'
)

plt.bar(
    top_rules['rule'],
    top_rules['confidence'],
    bottom=top_rules['support'],
    label='Confidence'
)
```

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```
Python 3.11.8 - C:\Users\Gazala\Documents\VSCode\workspace\mining.py (3.11.8)
File Edit Shell Debug Options Window Help
Python 3.11.8 (tags/v3.11.8:8206b45, Apr 8 2023, 14:47:33) [AMD64] on win32
Enter "help" below or click "Help" above for more information.
>>>
> SCRIPT: C:\Users\Gazala\Documents\workspace\mining.py
First five seconds:
Row ID      Order ID  Order Date  ...  Quantity  Discount  Profit
0      1  CA-2013-100998  08-11-2013  ...      2      0.00  41.9026
1      2  CA-2013-100998  08-11-2013  ...      2      0.00  219.8920
2      3  CA-2013-100998  08-11-2013  ...      2      0.00  6.8714
3      4  US-2013-100998  11-10-2012  ...      2      0.45 -385.9320
4      5  US-2013-100998  11-10-2012  ...      2      0.00  2.5164

[5 rows x 7 columns]

Transaction Data:
Sub-Category  Accessories  Appliances  Art  ...  Storage  Supplies  Tables
Order ID
CA-2013-100998  False      False      False  ...  False  False  False
CA-2013-100998  False      False      False  ...  False  False  True
CA-2013-100998  False      False      False  ...  False  False  False
CA-2013-100998  False      False      False  ...  False  False  False
CA-2013-100998  False      False      False  ...  False  False  False

[5 rows x 8 columns]

Frequent Itemsets:
support      itemsets
0  0.343982    frozenset({Accessories})
1  0.396098    frozenset({Appliances})
2  0.145307    frozenset({Art})
3  0.262727    frozenset({Binders})
4  0.044728    frozenset({Books})
...
43  0.023338    frozenset({Phones, Storage})
44  0.025991    frozenset({Tables, Phones})
45  0.010182    frozenset({Binders, Paper, Furnishings})
46  0.010781    frozenset({Binders, Phones, Paper})
47  0.010081    frozenset({Binders, Storage, Paper})

[68 rows x 2 columns]

Association Rules:
antecedents  ...  lift
27  frozenset({Phones, Paper})  ...  1.174494
29  frozenset({Storage, Paper})  ...  1.133320
9   frozenset({Fasteners})      ...  1.097611
2   frozenset({Appliances})     ...  1.097140
28  frozenset({Paper, Furnishings})  ...  1.094711
19  frozenset({Fasteners})       ...  1.104125
26  frozenset({Binders, Phones}) ...  1.141240
28  frozenset({Binders, Storage}) ...  1.120114
14  frozenset({Binders, Furnishings}) ...  1.053289
11  frozenset({Storage})         ...  0.974826

[10 rows x 3 columns]
```

Output:

```
Python 3.11.8 - C:\Users\Gazala\Documents\workspace\mining.py (3.11.8)
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